

L2b: Yield, Duration, Convexity, and the Yield Curve

CHEME 5660 Cornell University

Copyright Jeffrey Varner 2026. All Rights Reserved

Disclaimer and Risks

This material is for training and information only. It does not provide investment advice or recommend any security, derivative, trade, or strategy.

Trading involves risk. Past results do not guarantee future results. Any risk you take should fit your resources, experience, and tolerance for loss.

Your decisions are your responsibility. Base them on your own objectives, financial position, risk tolerance, and liquidity needs.

Scenario

Suppose you manage a $500 million Treasury portfolio. The Federal Reserve meets today, and the market expects a possible 50-basis-point move in the policy rate.

If yields move, the value of the portfolio will move with them, and your clients will want to know by how much.

The estimate depends on the timing of the portfolio's cash flows and on the change in each security's yield. We will use the slope and curvature of the price-yield curve to compute the price response.

Policy rate and yield: A half-point move in the *policy rate* does not imply a half-point move in *your security's yield*.

Learning Objectives

We start with the L2a pricing model and study how fixed-income prices respond when yields change.

Today's concepts

- Price and yield: Explain why a higher yield lowers the value of fixed positive cash flows and why the price-yield curve is convex.
- Duration and convexity: Use both measures to estimate a price change for a small change in yield, and state the units of each measure.
- Yield measures: Distinguish yield to maturity, par yield, spot rates, and maturity-specific discount factors.

Companion notebook: L2b lecture notebook

Lecture and Example Notebooks

Lecture: CHEME-5660-L2b-Lecture-Yield-Duration-Convexity-Curve-Fall-2026.ipynb

Example: CHEME-5660-L2b-Example-Treasury-Duration-Convexity-Fall-2026.ipynb

The example prices a coupon Treasury security and computes the price response to changes in yield and coupon rate.

The advanced/ directory contains optional notebooks on Malkiel's theorems and the yield curve.

Link: https://github.com/varnerlab/CHEME-5660-CourseRepository-Fall-2026.git

Treasury prices change when market yields change.

Are Treasury Securities Risk-Free?

Not exactly. U.S. Treasury securities have low default risk, but they are still exposed to interest-rate risk.

- Default risk: The issuer fails to make a promised payment. This risk is low for U.S. Treasury securities, although sovereign defaults do occur.
- Interest-rate risk: A change in market yields changes the value of fixed promised cash flows. The market value can fall even if the security is not sold.

Focus: We focus on interest-rate risk because we can compute how a change in yield changes the price.

Silicon Valley Bank and Interest-Rate Risk

Silicon Valley Bank held many long-maturity, fixed-rate government securities. Their market values fell when interest rates rose in 2022 and 2023.

- Many of the losses remained unrealized while the bank held the securities.
- In March 2023, uninsured depositors withdrew funds. To meet those withdrawals, SVB sold securities, realized losses, and announced a capital raise.
- The FDIC took control on March 10, 2023, when the bank had roughly $209 billion in assets. At the time, it was the second-largest failure of an insured bank on record; it is now the third-largest.

The decline in asset values and the deposit withdrawals reinforced each other. Interest-rate risk was important, but it was not the only cause of the failure.

Overview: the collapse of Silicon Valley Bank, from PBS

Malkiel's Bond-Pricing Theorems

Malkiel's five theorems describe the prices of option-free bonds with fixed positive cash flows. Each comparison holds the other contract terms fixed. The maturity comparisons use bonds priced at par, so $c = y$.

1. The price falls when the yield rises.
2. A longer-maturity bond is more sensitive to a change in yield.
3. Each additional period adds less sensitivity than the previous one.
4. A decrease in yield raises the price more than an equal increase in yield lowers it.
5. A higher coupon rate reduces the percentage price response to a given change in yield.

Reference: Malkiel (1962), *Expectations, Bond Prices, and the Term Structure of Interest Rates*, QJE 76(2), 197–218.

Treasury Bill Pricing

A Treasury bill has one cash flow: the face value V_P paid at maturity.

Let T be the remaining time to maturity in years, n the number of compounding periods per year, and y the nominal annual yield. From L2a, the price is:

$$V_B = \underbrace{(1 + y/n)^{-nT}}_{\mathcal{D}_{T,0}^{-1}(y;n)} V_P.$$

A larger yield or a longer maturity discounts the face value more heavily. We assume $V_P > 0$, $T > 0$, and $y > 0$ unless stated otherwise.

Price Sensitivity of a Treasury Bill

A bill has no coupon, so $c = 0$ and $dc = 0$. Holding V_P and n fixed, its price changes with the yield y and the remaining maturity T:

$$dV_B = -\underbrace{V_P \left(1 + y/n\right)^{-nT} \left(\frac{T}{1 + y/n}\right)}_{V_{B,y}} dy \; - \; \underbrace{V_P \left(1 + y/n\right)^{-nT} \left(n \log\left(1 + y/n\right)\right)}_{V_{B,T}} dT.$$

The quantities $V_{B,y}$ and $V_{B,T}$ are positive sensitivity magnitudes. The minus signs show that the price falls when either the yield or the remaining maturity increases, provided $y > 0$.

No dc term: The contract fixes $dc = 0$. The derivative $\partial V_B / \partial c$ is not zero, even at $c = 0$.

Interpreting a Change in Maturity

A change in T can describe either a comparison between contracts or the passage of time for one contract:

- Compare two contracts: A 6-month bill and a 12-month bill have different contractual maturities. For a coupon bond, a longer maturity can also add coupon payments.
- Hold one contract over time: As calendar time advances, the remaining maturity falls, so $dT = -dt$. For a coupon bond, this change is smooth only between coupon dates.

Treasury Bill Sensitivities

The sensitivity magnitudes for yield and maturity are:

$$V_{B,y} = \underbrace{V_P(1+y/n)^{-nT}}_{=V_B} \underbrace{\left(\frac{T}{1+y/n}\right)}_{>0}, \qquad V_{B,T} = \underbrace{V_P(1+y/n)^{-nT}}_{=V_B} \underbrace{(n\log(1+y/n))}_{=g_y},$$

where $g_y := n\log(1+y/n)$ is the continuously compounded rate equivalent to y.

- $V_{B,y} > 0$ for $T > 0$ and $1 + y/n > 0$.
- $V_{B,T} > 0$ requires $y > 0$.
- At $y = 0$, maturity does not affect price. For $-n < y < 0$, a longer bill has a higher price.

How Yield and Time Affect a Treasury Bill

Effect of a higher yield

A higher yield discounts the face value more heavily, so $y \uparrow \Rightarrow V_B \downarrow$. If a 52-week bill's yield rises from 2% to 3%, its market price falls.

Effect of time passing

At a fixed positive yield, the price rises as the bill approaches maturity because fewer discounting periods remain: $T \downarrow \Rightarrow V_B \uparrow$. At $y = 0$, the discount factor is one, so the price does not depend on maturity.

Optional Treasury Bill Example

The required lecture continues with coupon-bearing notes and bonds. The optional bill notebook examines maturity sensitivity in more detail.

Question: How does the price sensitivity of a zero-coupon bill change with its remaining maturity?

Notebook: The Treasury bill maturity example in advanced/malkiel/.

Coupon-Bearing Treasury Notes and Bonds

A coupon-bearing Treasury security pays coupons before maturity and returns the face value at maturity. Let c be the annual coupon rate, $C := (c/n)V_P$ the coupon paid each period, and $N := nT$ the number of remaining payment dates. The L2a pricing model is:

$$V_B = \mathcal{D}_{N,0}^{-1}(y;n)\,V_P + C\sum_{j=1}^{N}\mathcal{D}_{j,0}^{-1}(y;n), \qquad \mathcal{D}_{j,0}^{-1}(y;n) = (1+y/n)^{-j}.$$

The price has two parts:

- the discounted principal payment V_P at date N;
- the discounted coupon payments C at dates $j = 1, \ldots, N$.

Yield and Coupon Sensitivity

Hold the payment schedule fixed and differentiate:

$$\frac{\partial V_B}{\partial y} = -\frac{1}{n(1+y/n)}\left[NV_P\,\mathcal{D}^{-1}_{N,0}(y;n) + C\sum_{j=1}^{N} j\,\mathcal{D}^{-1}_{j,0}(y;n)\right] < 0,$$

$$\frac{\partial V_B}{\partial c} = \frac{V_P}{n}\sum_{j=1}^{N}\mathcal{D}^{-1}_{j,0}(y;n) > 0.$$

A higher yield discounts every payment more heavily, so the price falls. A higher coupon rate increases every coupon payment, so the price rises.

Schedule length: N is a discrete index. We do not differentiate a finite sum with upper limit nT as if T were continuous.

How Yield, Coupon, and Maturity Affect Price

- Yield: $y \uparrow \Rightarrow V_B \downarrow$ because every coupon and the principal payment are discounted more heavily.
- Coupon rate: $c \uparrow \Rightarrow V_B \uparrow$, but not in direct proportion. Doubling c doubles the coupon stream, not the principal payment.
- Remaining maturity: The direction of the price change depends on $c - y$. As maturity shortens, a discount bond ($c < y$) rises toward par, a premium bond ($c > y$) falls toward par, and a par bond ($c = y$) stays at par on coupon dates.

Maturity comparison: Malkiel's result concerns *sensitivity to a yield change*, not the price level.

A General Cash-Flow Pricing Model

For an option-free security, the promised cash flows do not change when yields change. Conventional noncallable Treasury securities satisfy this assumption.

Let $X_j \geq 0$ be the cash flow at period $j = 1, \ldots, N$, with at least one positive cash flow. The price is:

$$V_B(y) = \sum_{j=1}^{N} X_j \, \mathcal{D}_{j,0}^{-1}(y; n).$$

- Note or bond: $X_j = C$ for $j < N$ and $X_N = C + V_P$.
- Bill: $X_j = 0$ for $j < N$ and $X_N = V_P$.

Outside this model: callable, inflation-indexed, and mortgage-backed securities can have cash flows that change with rates or states.

The Price-Yield Relationship

Holding the cash flows X_j and the number of payments N fixed gives:

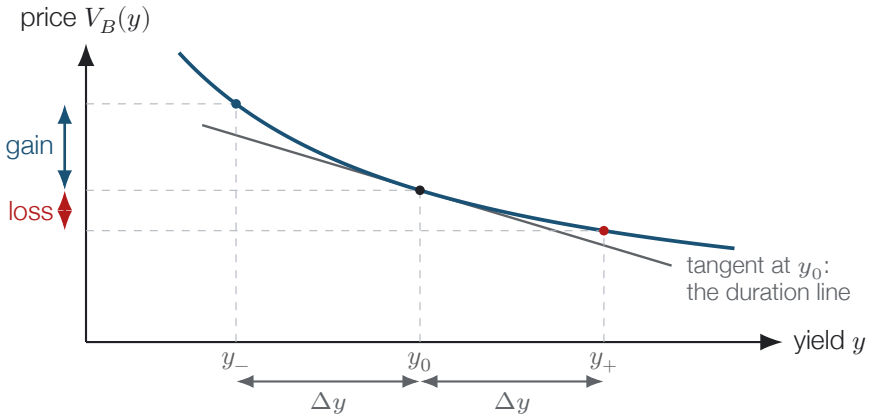
$$\frac{dV_B}{dy} = -\frac{1}{n}\sum_{j=1}^{N} j\, X_j \,(1+y/n)^{-j-1} < 0,$$

$$\frac{d^2V_B}{dy^2} = \frac{1}{n^2}\sum_{j=1}^{N} j(j+1) X_j \,(1+y/n)^{-j-2} > 0.$$

- $dV_B/dy < 0$: The price decreases as the yield increases.
- $d^2V_B/dy^2 > 0$: The price-yield curve is strictly convex.

Sign conditions: $1 + y/n > 0$, every $X_j \geq 0$, and at least one $X_j > 0$.

Convexity and Asymmetric Price Changes



For equal changes in yield, the gain from a decrease is larger than the loss from an increase. The tangent line gives the duration estimate, while convexity accounts for the curvature.

Modified Duration and Convexity

The price derivatives measure dollar sensitivity, so they depend on the size of the position. Dividing by the current price $V_B(y) > 0$ gives percentage sensitivity measures that can be compared across securities:

$$D_{\text{mod}}(y) := -\frac{1}{V_B(y)}\frac{dV_B}{dy}, \qquad K(y) := \frac{1}{V_B(y)}\frac{d^2V_B}{dy^2}.$$

- Modified duration D_{mod} measures the local percentage slope of price with respect to yield.
- Convexity K measures the local percentage curvature.
- If y is a decimal annual yield, D_{mod} has units of years and K has units of years squared.

Macaulay and Modified Duration

Macaulay duration is the present-value-weighted average time to payment. Modified duration converts that measure of cash-flow timing into local price sensitivity:

$$D_{\text{mod}} = \frac{D_{\text{mac}}}{1 + y/n}.$$

A bill makes its only payment at maturity, so:

$$D_{\text{mac}} = T, \qquad D_{\text{mod}} = \frac{T}{1 + y/n} = \frac{V_{B,y}}{V_B}.$$

Estimating a Price Change

For a small yield change Δy, the second-order Taylor approximation is:

$$\frac{\Delta V_B}{V_B} \approx \underbrace{-D_{\text{mod}}\,\Delta y}_{\text{slope}} + \underbrace{\tfrac{1}{2}K(\Delta y)^2}_{\text{curvature}}.$$

Because $K > 0$, the convexity term is positive for either sign of Δy. It increases the estimated gain when the yield falls and reduces the estimated loss when the yield rises.

Use decimal yield changes: one basis point is 10^{-4}, so 50 basis points is $\Delta y = 0.0050$.

Policy rate and yield: A 50-basis-point move in the *policy rate* does not imply $\Delta y = 0.0050$ for a given security.

Example: A 50-Basis-Point Yield Increase

Suppose a note has $D_{\text{mod}} = 7.2$ years and $K = 65$ years^2. If its yield rises by 50 basis points, then $\Delta y = 0.0050$:

- The duration term is $-7.2 \times 0.0050 = -0.0360$.
- The convexity term is $\frac{1}{2} \times 65 \times (0.0050)^2 = +0.0008$.
- The estimated price change is -3.52%, compared with -3.60% using duration alone.

Duration and convexity are local measures. For a large yield change, reprice the full cash-flow schedule at the new yield.

Portfolio Duration

- Duration places securities with different coupons, maturities, and prices on the same sensitivity scale.
- Portfolio duration is the market-value-weighted average of the durations of the individual positions.
- Offsetting positions can reduce net duration. Funding long-duration assets with short-duration liabilities creates a duration mismatch.

These calculations assume that each position experiences the stated yield change. If different maturities move by different amounts, we need maturity-specific yield changes.

Duration of a Par Bond

To compare sensitivity across maturities, consider option-free bonds with fixed positive cash flows and hold the other terms fixed. Each bond starts at par, so $c = y$.

Sum the coupon-bond price, differentiate at fixed c, and then set $c = y$:

$$\frac{V_B}{V_P} = \frac{c}{y} + \left(1 - \frac{c}{y}\right)(1 + y/n)^{-N} \qquad \Longrightarrow \qquad D_{\text{mod}}(N) = \frac{1 - (1 + y/n)^{-N}}{y}.$$

The order of these steps matters. If we set $c = y$ before differentiating, the price remains at V_P along the par curve and the derivative is zero. We assume $y > 0$.

Malkiel's Theorems 1 and 2

1. The price falls when the yield rises

The result $dV_B/dy < 0$, or equivalently $D_{\text{mod}} > 0$, follows because a higher yield discounts every positive payment more heavily.

2. A longer par bond is more sensitive to yield

$$D_{\text{mod}}(N) = \frac{1 - (1 + y/n)^{-N}}{y}$$

increases with N. Because each bond starts at $V_B = V_P$, the dollar sensitivity $-dV_B/dy = V_P D_{\text{mod}}(N)$ also increases with maturity.

Malkiel's Theorem 3

Adding one payment period changes modified duration by:

$$D_{\text{mod}}(N+1) - D_{\text{mod}}(N) = \frac{(1+y/n)^{-N-1}}{n} > 0.$$

The next increment equals the previous increment multiplied by:

$$(1+y/n)^{-1} < 1.$$

Modified duration therefore increases with maturity, but each additional period adds less than the previous one. As $N \to \infty$, modified duration approaches $1/y$.

Limit: $1/y$ is the modified duration of a perpetuity.

Malkiel's Theorem 4

Start at the same yield and compare equal changes $\pm\Delta y$.

$$\frac{\Delta V_B}{V_B} \approx -D_{\text{mod}}\Delta y + \frac{1}{2}K(\Delta y)^2$$

The duration term changes sign when the direction of the yield change changes. The positive convexity term does not. Therefore, a yield decrease raises the price more than an equal yield increase lowers it.

The direction of the curvature matters. A concave price-yield curve would produce the opposite asymmetry.

Malkiel's Theorem 5

A higher coupon shifts more of the bond's present value to earlier payment dates. This lowers Macaulay duration and therefore lowers D_{mod}.

This comparison requires $N \geq 2$ and concerns percentage sensitivity. Dollar sensitivity can still increase because a higher coupon raises the bond's price:

$$-\frac{dV_B}{dy} = V_B D_{\text{mod}}.$$

Edge case: at $N = 1$, every payment arrives on one date and $D_{\text{mod}} = 1/[n(1 + y/n)]$, regardless of coupon size.

Why Theorems 2 and 3 Compare Bonds at Par

Away from par, percentage sensitivity and dollar sensitivity can rank bonds differently. For a zero-coupon bill, $D_{\text{mod}} = T/(1 + y/n)$ continues to grow with maturity, but dollar sensitivity contains one factor that grows and another that shrinks:

$$-\frac{dV_B}{dy} = V_P \underbrace{\frac{T}{1 + y/n}}_{\text{growing}} \underbrace{(1 + y/n)^{-nT}}_{\text{shrinking}}.$$

With $V_P = 100$, $n = 1$, and $y = 5\%$, dollar sensitivity increases at first, reaches a maximum near $T \approx 20.5$ years, and then decreases. It is about 718 USD per unit of yield at the maximum and about 541 at $T = 40$.

The 40-year bill is more sensitive in percentage terms but less sensitive in dollars. Starting both bonds at par makes the rankings agree.

Example: CHEME-5660-L2b-Example-Treasury-Duration-Convexity-Fall-2026.ipynb, which tests Theorems 4 and 5.

Yield Measures and the Yield Curve

- Yield to maturity is the single rate that reproduces a security's market price from its promised cash flows.
- A spot rate applies to one payment date, so the spot curve provides a separate discount factor for each date.
- A par yield is the coupon rate that prices a hypothetical bond at face value under the spot curve. The published Treasury yield curve is a par yield curve.

Let $D(0, t_j)$ be the time-0 discount factor for a payment at date t_j. Then:

$$V_B = \sum_{j=1}^{N} X_j \, D(0, t_j).$$

The level, slope, and curvature of the yield curve describe how discounting varies with maturity. They do not, by themselves, determine future short rates.

Inverted Yield Curves

A yield curve is inverted when a long-maturity yield is below a short-maturity yield.

An inverted spread such as 10-year minus 3-month has preceded every U.S. recession of the past half century.

This spread does not tell us when a recession will begin. False positives occur, lead times range from months to years, and the result depends on the maturity pair. The optional par-yield example computes the spread from Treasury data.

Notebook: advanced/yield_curve/CHEME-5660-L3a-ParYieldCurves-Example-Fall-2026.ipynb

Optional Advanced Material

Malkiel: advanced/malkiel/
The Malkiel notebooks study Treasury-bill maturity data and derive a closed-form proof of Theorems 2 and 3.

Yield curve: advanced/yield_curve/
The yield-curve notebooks compute investment yield, solve for yield to maturity with the secant method, and construct par yield curves.

STRIPS: advanced/strips/
The STRIPS notebooks infer spot, par, and implied forward rates from Treasury STRIPS prices.

These notebooks are optional and are not prerequisites for L3a.

Notebooks: lectures/week-2/L2b/advanced, also linked from the lecture notebook.

Summary

We used cash-flow pricing to measure the interest-rate risk of fixed-income securities.

Key takeaways

- **Price and yield move in opposite directions.** For fixed positive cash flows, the price-yield curve is decreasing and convex.
- **Duration and convexity estimate small price changes.** The sensitivity depends on the maturity, coupon, and current yield.
- **Different yield measures answer different questions.** Yield to maturity summarizes one security, spot rates price individual payment dates, and par yields summarize the spot curve as coupon rates.

How can duration and convexity help us manage the interest-rate risk of a bond portfolio?